

Aerial Multi-Object & Multi-Camera Tracking System



Multi-Camera

Real-time Tracking

AI-Powered

Project Overview



Real-time Detection & Tracking

Advanced object detection and tracking system operating in real-time



Multi-Object & Multi-Camera Support

Simultaneously tracks multiple objects across multiple camera streams

Aerial Robotics Ready

Designed for drone follow-me systems and robot vision research



Intelligent Recovery

Auto-recenter and auto-relock capabilities for robust tracking



Comprehensive Analytics

Distance, velocity, angle measurements with CSV logs and graphs

Technical Architecture



Multi-Camera Inputs



YOLOv8 Detection



DeepSORT Tracking



Kalman Filter



PID Gimbal Align



Distance & Velocity



CSV Logger



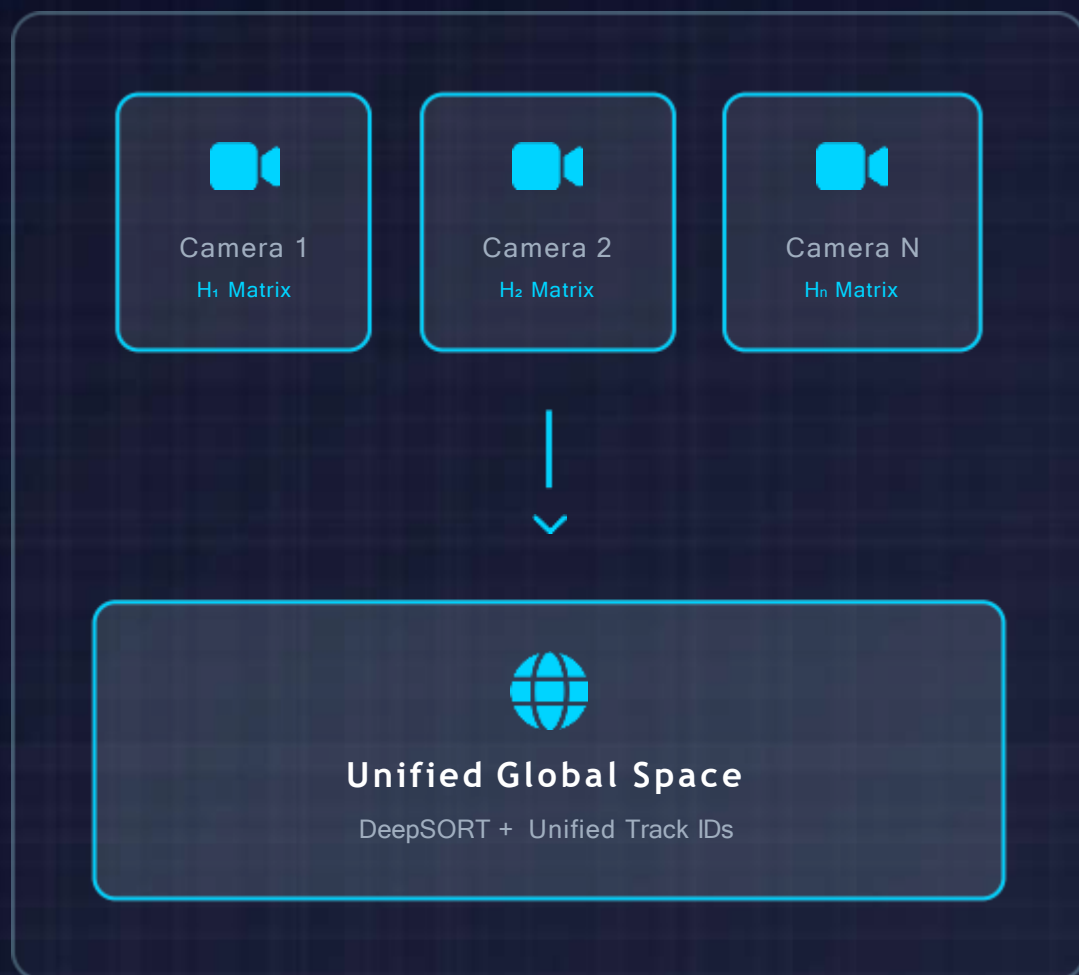
Graph Generator



Multi-Cam Fusion

Homography Projection

Multi-Camera Fusion



Fusion Process

- Each camera has a 3×3 homography mapping image $\rightarrow$ world plane
- All detections projected into a **single global space**
- DeepSORT runs globally $\rightarrow$ unified track IDs

Benefits

Larger effective FOV

Fewer blind spots

Better continuity

Robust tracking

Tech Stack



Python 3.10



YOLOv8
(Ultralytics)



DeepSORT
ReID Tracking



OpenCV
Video I/O, Tracking



NumPy
Calculations



Matplotlib
Graph Generation



Homography
Multi-Camera Fusion



CSV Logging
+ Analytics



Modular Architecture

Easy to extend for drones, ROS, or MAVSDK

Challenges & Problems Faced



Integrating YOLO, DeepSORT, Kalman, PID

Complex integration of multiple algorithms and ensuring seamless data flow between components

Ensuring Real-Time Performance

Optimizing computational efficiency to maintain high frame rates for aerial tracking applications

Handling Occlusions & ID Switching

Maintaining consistent object IDs when targets are temporarily occluded or cross paths

Implementing Auto-Recenter & Auto-Relock

Developing robust mechanisms to automatically recenter and relock on targets during dynamic movements

Advantages & Use Cases

★ Advantages

-  Highly modular & scalable
-  Tracks multiple objects across multiple cameras
-  Smooth tracking even with occlusions
-  Real-world metrics: distance, velocity, angle
-  Global tracking IDs across cameras
-  Ideal for robotics experiments

💡 Use Cases

-  Autonomous drones (follow-me, surveillance)
-  Multi-camera security systems
-  Aerial robotics research
-  Traffic tracking (vehicles, pedestrians)
-  Sports analytics with multiple camera views



Thank You

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